

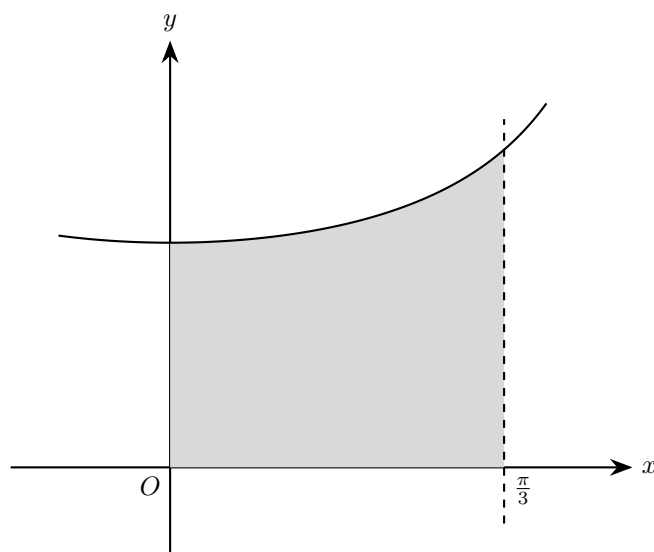
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1. (a) Show that

$$\frac{d}{dx} \ln(\sec x + \tan x) = \sec x$$

[2]

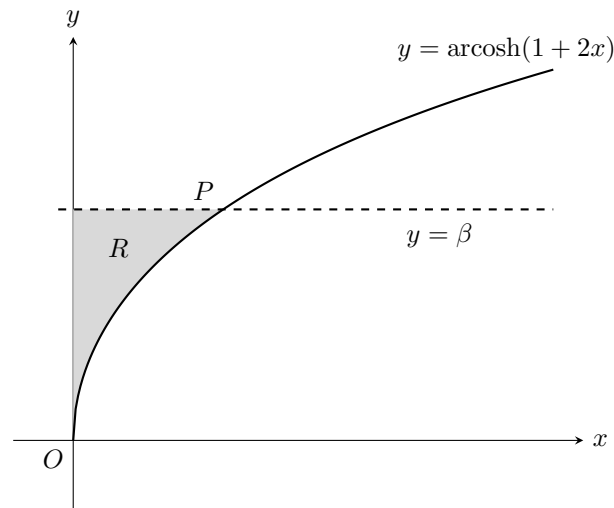


The shaded region is enclosed by the curve $y = \sqrt{\sec x}$, the coordinate axes and the line $x = \frac{\pi}{3}$.

The region is rotated completely about the x -axis.

(b) Find the exact volume of the solid generated.

[3]



2. The diagram above shows a sketch of part of the curve with equation

$$y = \operatorname{arcosh}(1 + 2x) \quad x \geq 0$$

and the straight line with equation $y = \beta$.

The line and the curve intersect at the point with coordinates (α, β) .

Given that $\beta = \ln(2 + \sqrt{3})$,

(a) show that $\alpha = \frac{1}{2}$. [3]

The finite region R , shown shaded in the diagram above, is bounded by the curve with equation $y = \operatorname{arcosh}(1 + 2x)$, the y -axis and the line with equation $y = \beta$.

The region R is rotated through 2π radians about the y -axis.

(b) Use calculus to find the exact value of the volume of the solid generated. [6]

3. The equation of a curve is

$$y = \sqrt{\frac{k-x}{k+x}}$$

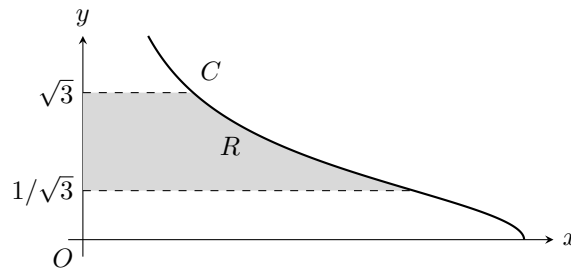
where k is a positive constant. The region enclosed by the curve, the x -axis, the y -axis and the line $x = k$ is rotated through 2π radians about the x -axis.

Given that the volume of the solid of revolution formed is 1 unit^3 , find the exact value of k . [4]

4. (a) Find

$$\int \sin^2 \theta \, d\theta$$

[2]



The diagram shows part of the curve C with parametric equations $x = 6 \sin^2 \theta$, $y = \cot \theta$, $0 < \theta \leq \frac{\pi}{2}$.

The finite region R shown in the diagram is bounded by C , the lines $y = \frac{1}{\sqrt{3}}$, $y = \sqrt{3}$ and the y -axis. Region R is rotated through 2π radians about the y -axis to form a solid of revolution.

(b) Show that the volume of the solid of revolution formed is given by the integral

$$k \int_a^b \sin^2 \theta \, d\theta$$

where a , b and k are constants to be found.

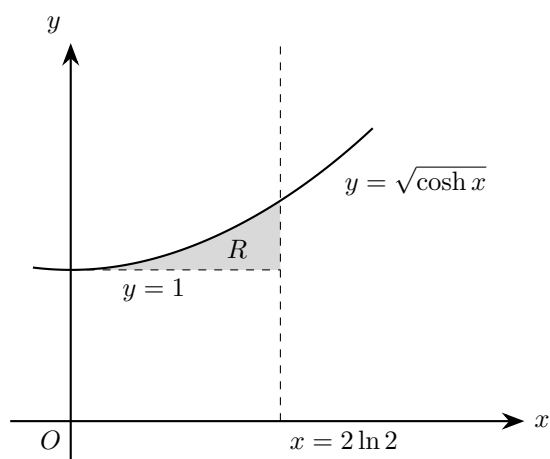
[5]

(c) Hence find the exact value of this volume, giving your answer in the form $p\pi^2$, where p is a constant to be found.

[3]

5. (a) Show that $\sinh(2 \ln 2) = \frac{15}{8}$

[2]

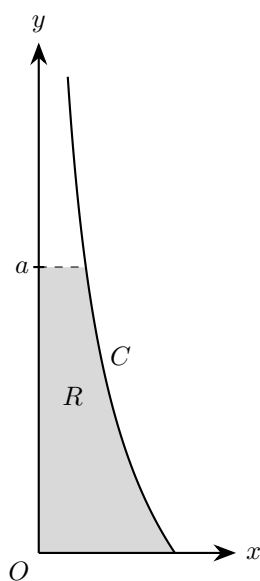


The region R is bounded by the curve with equation $y = \sqrt{\cosh x}$, the line $y = 1$ and the line $x = 2 \ln 2$, as shown in the diagram. The units of the axes are centimetres.

A designer models a hollow glass ornament as the solid formed by rotating R completely about the x -axis.

(b) Determine, according to the model, the exact volume of the ornament.

[4]



6. The curve C is given parametrically by

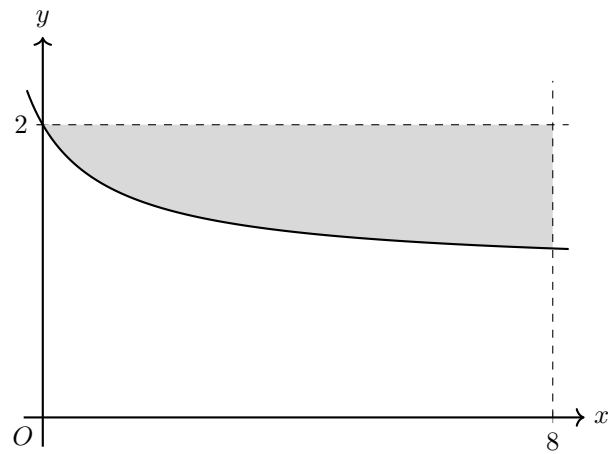
$$x = \frac{9}{t^4}, \quad y = t^2 - 3, \quad t \geq \sqrt{3}$$

The shaded region R , shown in the diagram, is enclosed by C , the x -axis, the y -axis and the line $y = a$. When R is rotated through 2π radians about the y -axis, the resulting solid has volume

$$\frac{7\pi}{8}$$

Find the exact value of a .

[8]



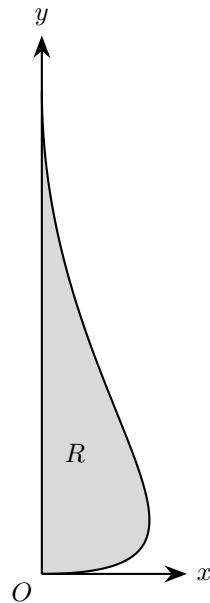
7. The diagram shows the region bounded by the curve $y = \sqrt{\frac{x+4}{x+1}}$, the line $y = 2$ and the line $x = 8$. The curve meets the line $y = 2$ at the point $(0, 2)$.

Find the exact volume of the solid formed when this region is rotated through 360° about the x -axis.

[6]

8. The region in the first quadrant bounded by the curve $y = \sinh x$, the y -axis, and the line $y = 2$ is rotated through 360° about the x -axis.

Find the exact volume of revolution generated, expressing your answer in a form involving a logarithm. [7]



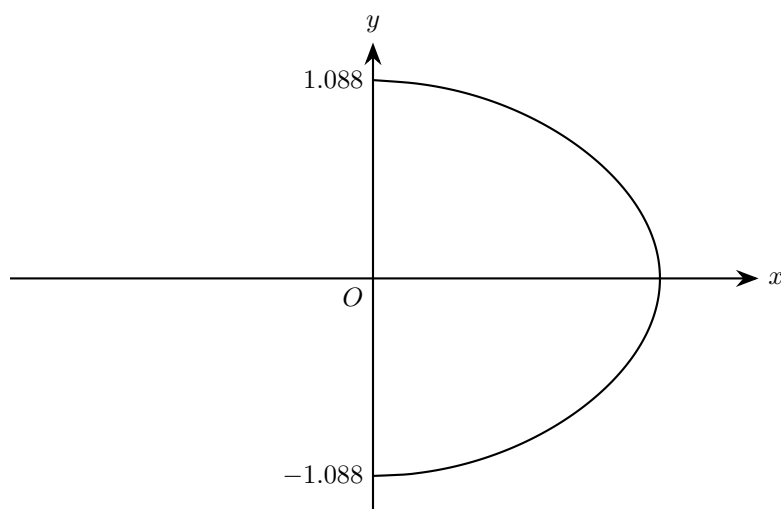
9. Part of the side profile of a solid wooden ornament is shown in the diagram.
The outline is modelled by the curve with parametric equations

$$x = 12t(1 - t)^2, \quad y = 8t^2, \quad 0 \leq t \leq 1$$

The ornament is formed by rotating the shaded region through 2π radians about the y -axis.

Use the model to find the volume of the ornament.

[7]



10. A chocolatier makes hand-rolled chocolate drops.

The tallest drop in one tray was approximately 2.2 cm high.

The shape of this drop is modelled by rotating the curve with equation

$$20x^2 + 6y^2 + y \sin(2y) = 8, \quad x \geq 0$$

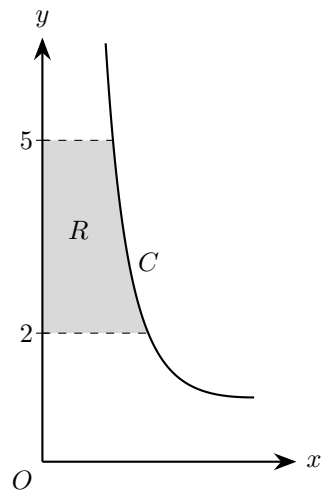
shown in the diagram above, about the y -axis through 2π radians, where the units are cm.

Given that the y -intercepts of the curve are -1.088 and 1.088 to four significant figures,

- (a) Use algebraic integration to determine, according to the model, the volume of this chocolate drop. [6]

The chocolatier melts down 80 chocolate drops from the same tray.

- (b) Use your answer to part (a) to decide whether, in reality, there is likely to be enough chocolate to fill a mould of volume 140 cm^3 , giving a reason. [2]



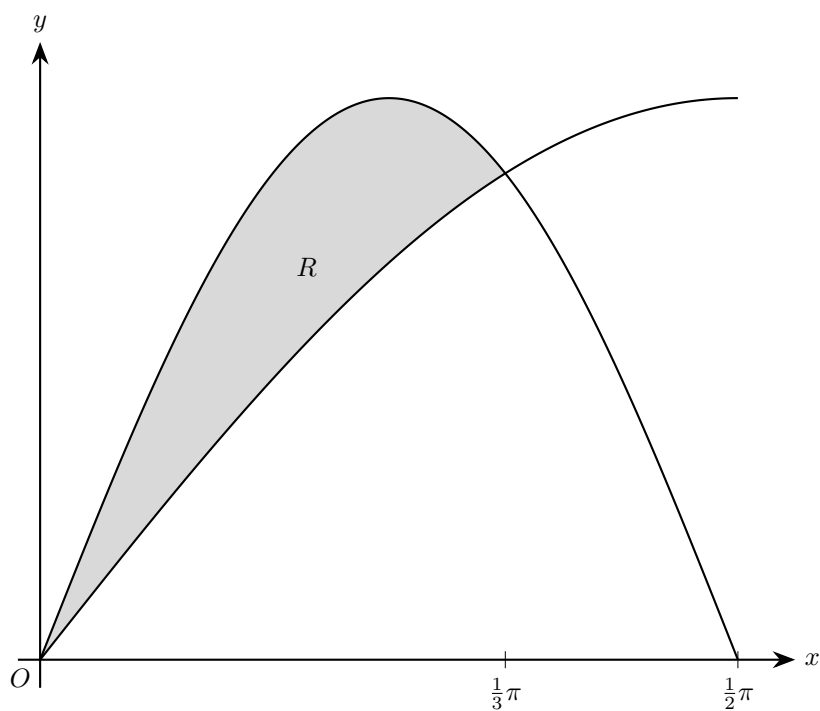
11. The diagram shows part of the curve C with parametric equations

$$x = \frac{2}{1+t}, \quad y = t^2 + 1, \quad t \geq 0$$

The region R is bounded by the curve, the y -axis and the lines $y = 2$ and $y = 5$. Region R is rotated through 2π radians about the y -axis.

Use parametric integration to find the volume of the resulting solid of revolution.

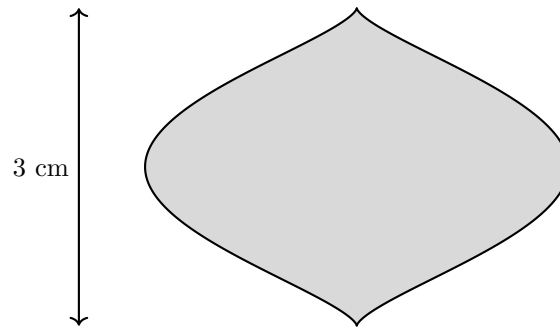
[6]



- 12.** The diagram shows the curves $y = \sin x$ and $y = \sin 2x$, for $0 \leq x \leq \frac{1}{2}\pi$. The shaded region R is the finite region enclosed by the curves.

Find the volume of the solid of revolution formed when R is rotated completely about the x -axis, giving your answer in terms of π .

[7]



- 13.** The diagram shows the outline of a glass bead. The bead is modelled by the solid obtained when the region enclosed by a curve C is rotated about the y -axis. The actual bead has height 3 cm. The curve C has parametric equations

$$x = \sin \theta - \frac{1}{3} \sin 3\theta, \quad y = 1 - \cos \theta, \quad 0 \leq \theta \leq 2\pi$$

- (a)** Show that a Cartesian equation of the curve C is

$$x^2 = \frac{16}{9}y^3(2-y)^3 \quad [4]$$

- (b)** Hence, using the model, find, in cm^3 , the volume of the bead. [5]

14. The equation of a curve is

$$y = \frac{x}{\sqrt{k^2 + x^2}}$$

where k is a positive constant.

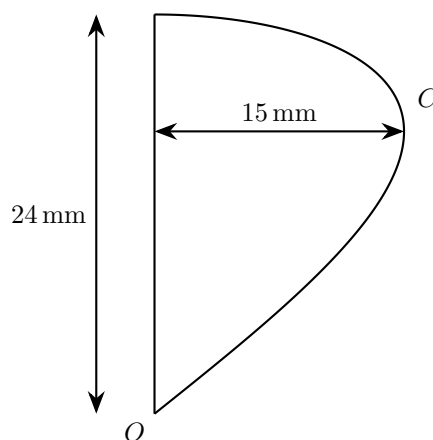
The region in the first quadrant bounded by the curve, the x -axis, the y -axis and the line $x = k$ is rotated through 2π radians about the x -axis

Given that the volume of revolution formed is 1 unit³, find the exact value of k

[4]

15. (a) Prove that

$$\cos 4\theta \cos \theta \equiv \frac{1}{2}(\cos 5\theta + \cos 3\theta) \quad [3]$$



The diagram shows the cross-section of a decorative ornament.

The ornament can be modelled by rotating the region enclosed by the curve C and the y -axis about the y -axis. Curve C has parametric equations

$$x = 15 \sin 2\theta, \quad y = 24 \sin \theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

where x and y are measured in millimetres.

(b) Find the volume of the ornament. [5]